Lost in time without an anchor: cerebellar dysfunction distorts cortical neural dynamics of subjective temporal judgments

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Background and questions

The human cerebellum is crucial for interval timing [1,2].

Two modes of interval timing

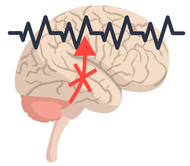
Explicit timing
Estimating and judging temporal intervals

Implicit timing
Using interval regularities to optimize behavior

Previous work with cerebellar ataxia (CA) patients in implicit timing: cerebellar degeneration abolishes interval-based adjustment of anticipatory cortical neural dynamics [3], pointing to cerebellar control.

Q1: *How* does the cerebellum impact cortical dynamics of explicit timing?

Q2: *At which stage* of the temporal information processing?



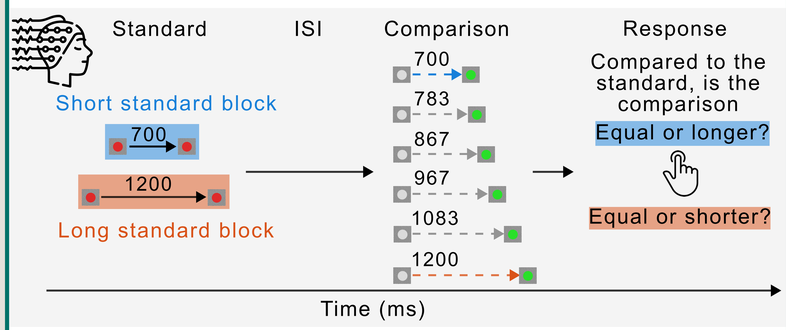
Temporal estimation

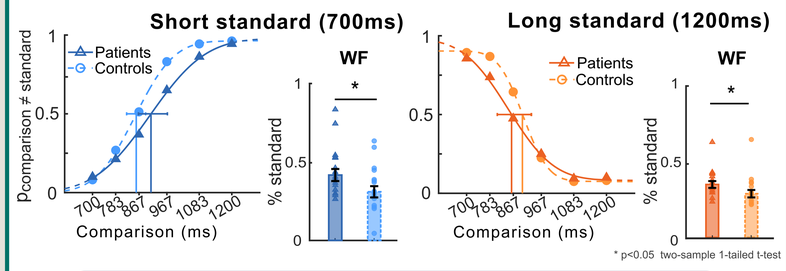
Temporal decision-making

References
[1] Ivry, R., & Schlerf, J., (2008) [2] Koch, G., Oliveri, M. & Caltagirone, C. (2009) [3] Breska, A., & Ivry, R., (2020)

Task design

$N_{CA\ patients}$ = 16, $N_{controls}$ = 16



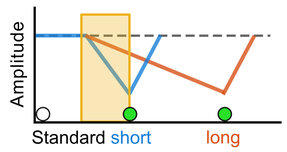


☞ Lower temporal perceptual sensitivity in patients compared to age-matched healthy controls

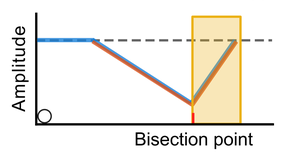
EEG markers of temporal decision-making

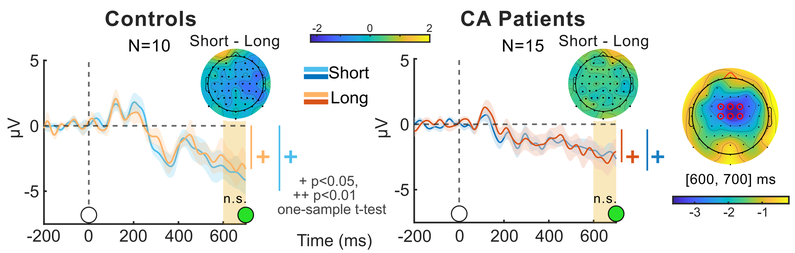
Comparison
CNV delta ITPC (0.54 - 2.18 Hz)

Standard-based strategy

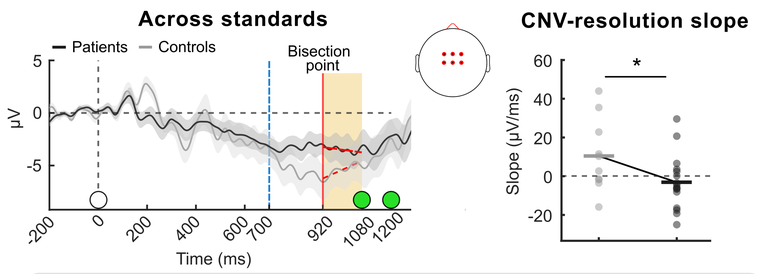


Anchor-based strategy

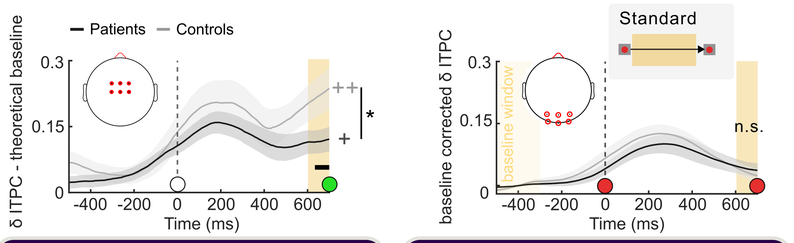




☞ No standard-based CNV adjustment in both groups



CNV resolution consistent with anchor-based model in controls, abolished in patients ☞ **causal role of the cerebellum in timing an internally generated anchor**

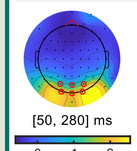


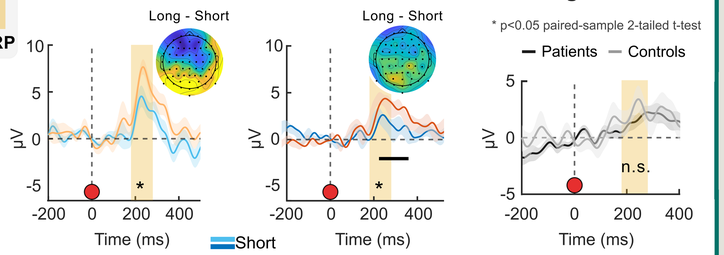
Higher temporal variability in patients at earliest comparison ☞**distinct functional roles of CNV and delta ITPC**

Comparable temporal variability across groups at standard onset ☞**distorted dynamics specific to comparison timing only**

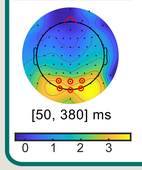
Preserved standard encoding and VEPs in patients

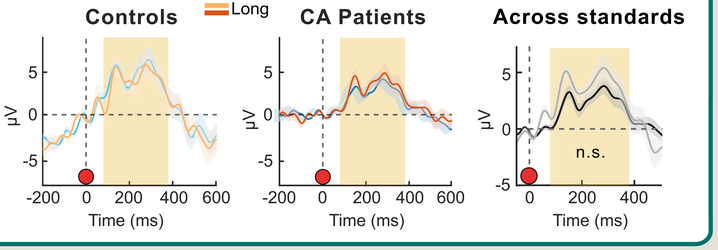
Standard
Post-offset ERP



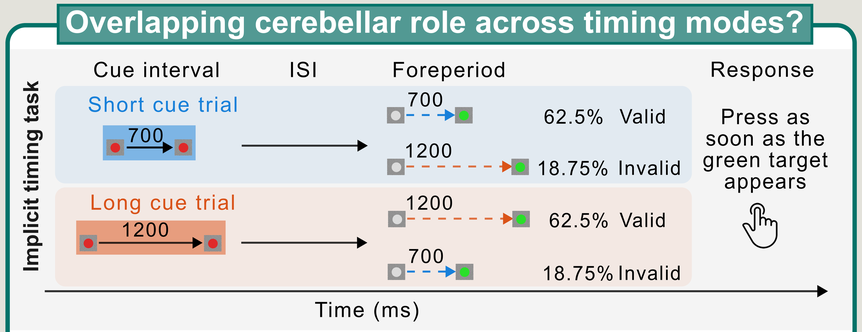


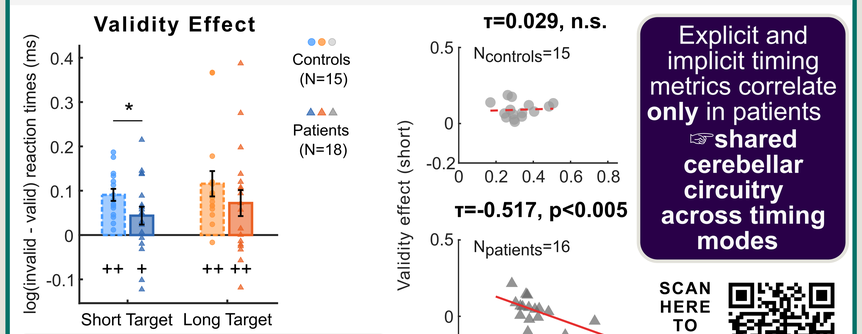
Standard
Visual Evoked Potential (VEP)





Overlapping cerebellar role across timing modes?





☞Reduced cue-validity effect in temporal orienting in patients

Explicit and implicit timing metrics correlate **only** in patients ☞**shared cerebellar circuitry across timing modes**

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